Getting Started

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Instructions

This document runs a simple analysis of the Table 1 from (Wansink and Payne 2009). Edit the header information to show your name and the date you complete the assignment.

Modify this document to analyze either Calories per Serving or Servings per Recipe. Document any changes you make in the literate portion of the file. Comment on your choice of measure to analyze.

Change the name of this file to match your user name on D2L, keeping the 'Rmd' extension, and include week number in the title (for example, `Peter.Claussen.1.Rmd`). Upload this file to D2L. Typeset this file to Word or PDF and upload the result to D2L as well.

Data

Table 1: Mean and (SD) for selected recipes from "Joy of Cooking"

Measure	1936	1946	1951	1963	1975	1997	2006
calories per	2123.8	2122.3	2089.9	2250.0	2234.2	2249.6	3051.9
recipe (SD)	(1050.0)	(1002.3)	(1009.6)	(1078.6)	(1089.2)	(1094.8)	(1496.2)
calories per	268.1	271.1	280.9	294.7	285.6	288.6	384.4
serving (SD)	(124.8)	(124.2)	(116.2)	(117.7)	(118.3)	(122.0)	(168.3)
servings per	12.9	12.9	13.0	12.7	12.4	12.4	12.7
recipe (SD)	(13.3)	(13.3)	(14.5)	(14.6)	(14.3)	(14.3)	(13.0)

Analysis

Enter data

```
CookingTooMuch.dat <- data.frame(
  Year=c(1936, 1946, 1951, 1963, 1975, 1997, 2006),
  CaloriesPerRecipeMean = c(2123.8, 2122.3, 2089.9, 2250.0, 2234.2, 2249.6, 3051.9),
  CaloriesPerRecipeSD = c(1050.0, 1002.3, 1009.6, 1078.6, 1089.2, 1094.8, 1496.2),
  CaloriesPerServingMean = c(268.1, 271.1, 280.9, 294.7, 285.6, 288.6, 384.4),
  CaloriesPerServingSD = c(124.8, 124.2, 116.2, 117.7, 118.3, 122.0, 168.3),
  ServingsPerRecipeMean = c(12.9, 12.9, 13.0, 12.7, 12.4, 12.4, 12.7),
  ServingsPerRecipeSD = c(13.3, 13.3, 14.5, 14.6, 14.3, 14.3, 13.0)
)
```

Create values for confidence interval plot

Wansink reports that 18 recipes were analyzed.

```r
n <- 18
```

Assume a significance level α of 5%.

```r
alpha <- 0.05
```

Use standard formula for standard error $\sigma/\sqrt{n}$ and confidence interval $t_{\alpha/2} \times s.e.$.

```r
StandardError <- function(sigma, n) {
  sigma/sqrt(n)
}
ConfidenceInterval <- function(sigma, n) {
  qt(1-alpha/2, Inf)*StandardError(sigma,n)
}
```

Create a variable for plotting and calculate upper and lower bounds using confidence intervals. For this assignment, I am plotting ServingsPerRecipe.

```r
PlotCookingTooMuch.dat <- CookingTooMuch.dat
PlotCookingTooMuch.dat$ServingsPerRecipe <-
  PlotCookingTooMuch.dat$ServingsPerRecipeMean
PlotCookingTooMuch.dat$Lower <-
  PlotCookingTooMuch.dat$ServingsPerRecipe - ConfidenceInterval(CookingTooMuch.dat$ServingsPerRecipeSD,
PlotCookingTooMuch.dat$Upper <-
  PlotCookingTooMuch.dat$ServingsPerRecipe + ConfidenceInterval(CookingTooMuch.dat$ServingsPerRecipeSD,
```

Here, we are selecting only ServingsPerRecipe variable for plotting.

```r
PlotCookingTooMuch.dat <-
  PlotCookingTooMuch.dat[, c("Year", "ServingsPerRecipe", "Lower", "Upper")]
```

Examine the values to make sure we've entered correctly.

```r
print(PlotCookingTooMuch.dat)
```

```
##   Year ServingsPerRecipe    Lower    Upper
## 1 1936              12.9 6.755826 19.04417
## 2 1946              12.9 6.755826 19.04417
## 3 1951              13.0 6.301465 19.69854
## 4 1963              12.7 5.955268 19.44473
## 5 1975              12.4 5.793858 19.00614
## 6 1997              12.4 5.793858 19.00614
## 7 2006              12.7 6.694417 18.70558
```

Couldn't find the confidence intervals for ServingsPerRecipe in Wanskins report for 1936 and 2006. So I am using the Reference CI as calculated CI using confidence interval function above. So here, I am using ComValues as ReferneceValues

```r
CompValues <- PlotCookingTooMuch.dat[c(1, 7), c("Lower", "Upper")]
#ReferenceValues <- matrix(c(1638.7, 2608.9, 2360.7, 3743.1),nrow=2,byrow=TRUE)
ReferenceValues <- CompValues
CompValues
```

```
##      Lower    Upper
## 1 6.755826 19.04417
## 7 6.694417 18.70558
```

```
ReferenceValues
```

```
##      Lower    Upper
## 1 6.755826 19.04417
## 7 6.694417 18.70558
```

```
any(abs(CompValues - ReferenceValues) > 0.1)
```

```
## [1] FALSE
```

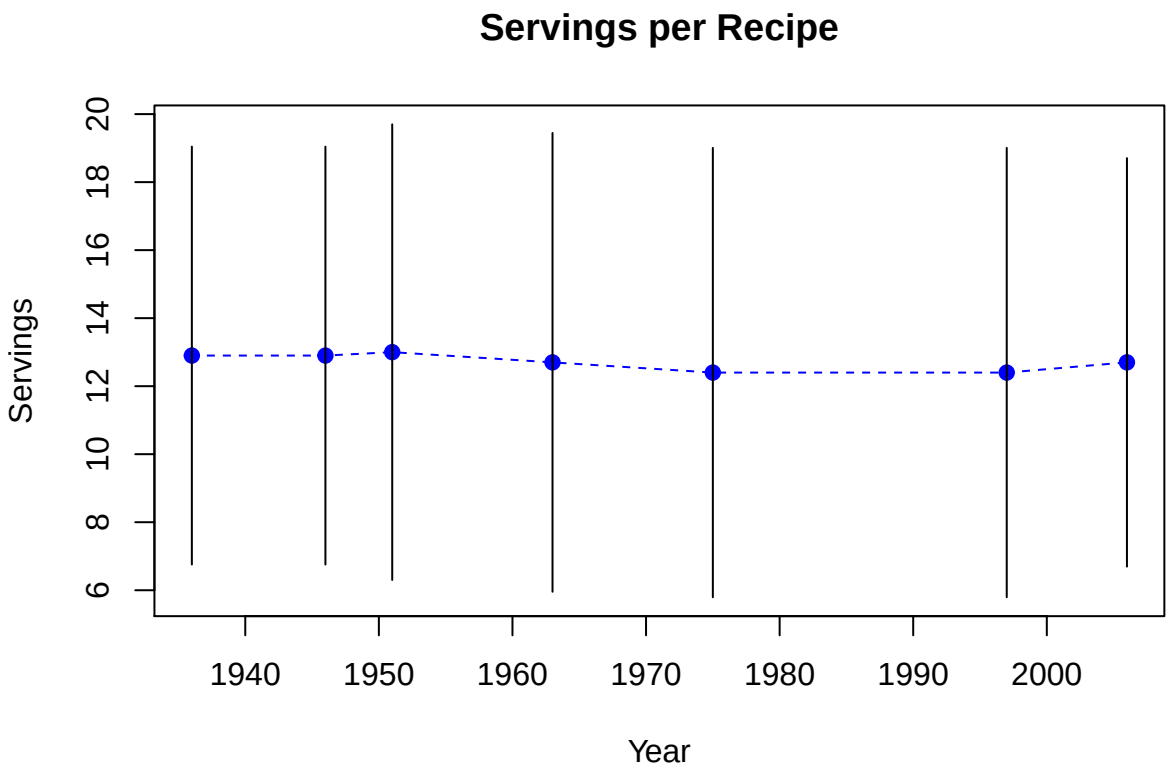
We no longer need the original data.

```
CookingTooMuch.dat <- NULL
```

Plot the table

#Here, changed the title and y and x labels to Servings Per Recipe

```
plot(
  ServingsPerRecipe ~ Year,
  data = PlotCookingTooMuch.dat,
  col = "blue",
  pch = 19,
  main = "Servings per Recipe",
  ylab = "Servings",
  ylim = c(
    min(PlotCookingTooMuch.dat$Lower),
    max(PlotCookingTooMuch.dat$Upper)
  )
)
lines(
  ServingsPerRecipe ~ Year,
  data = PlotCookingTooMuch.dat,
  lty = "dashed",
  col = "blue",
  lend = 2
)
segments(
  x0 = PlotCookingTooMuch.dat$Year,
  y0 = PlotCookingTooMuch.dat$Lower,
  x1 = PlotCookingTooMuch.dat$Year,
  y1 = PlotCookingTooMuch.dat$Upper
)
```



Comments

From this plot, it appears that average servings per recipe doesn't seem to change from 1936 to 2006.

References

Wansink, Brian, and Collin Payne. 2009. "The Joy of Cooking Too Much: 70 Years of Calorie Increases in Classic Recipes" 150 (150): 291–92.